

628 runs, 0 failures (incl. 60 ViT-B/16 completing B1’s third backbone) • dsisco01 **results/track_b/** • answers the B1–B8 items of **paper/HANDOFF_TRACK_B.tex** • all gaps matched-seed (paired) per cell; backbones never pooled. “Best baseline” = the strongest method that *trains* to satisfy the cap (Fioretto / ALM / Hounie); post-hoc clipping is a separate category (below).

Bottom line. TraLO’s supported win is against the **best trained baselines** (the constrained-optimization duals) on OctMNIST tight caps: +0.028 **cc-F1 vs the strongest dual (ALM)** and +0.039 **vs Fioretto** (6/6 cells; bootstrap CIs exclude 0 at L30/L40), every backbone positive, 4 backbones. **Post-hoc clipping is a separate category:** it ties TraLO on cc-F1 but does *not* satisfy the cap natively (TraLO 1.0 vs clip 0.0). **Focal** ties TraLO’s macro-F1 on Derm. **Mechanism:** the count penalty re-ranks the boundary only while CE is unsaturated (headroom) – this explains every win and every tie.

▷ **Status for this presentation:** every result below is **final and concluded**. B1’s third backbone (ViT-B/16) has now landed (60/60 runs, 0 failed) and its rows are folded into the B1 table; the three-backbone coverage confirms the tie-on-quality verdict without changing the story. Sections B2–B8 were already complete.

B1 Does an imbalanced clipper (focal / class-bal / logit-adjust) close TraLO’s macro-F1 gap?

312 runs: 5 methods × Oct/Derm/Tissue × MNV3/RegNet/ViT-B/16 × warm-up{1,5,50} × L30 × 4 seeds. Table = macro-F1 gap TraLO – baseline at warm-up 50 (positive = TraLO better).

dataset	backbone	vs focal	vs class-bal	vs logit-adj	vs clip
Derm	MNV3	−0.012	−0.016	+0.021	+0.010
Derm	RegNet	+0.006	+0.022	+0.042	+0.013
Derm	ViT	−0.020	−0.017	+0.025	+0.009
Oct	MNV3	+0.025	+0.023	+0.021	+0.023
Oct	RegNet	−0.005	+0.022	−0.000	+0.022
Oct	ViT	−0.004	+0.010	−0.006	−0.001
Tissue	MNV3	+0.020	+0.014	+0.026	+0.029
Tissue	RegNet	−0.002	+0.003	−0.001	+0.001
Tissue	ViT	+0.017	+0.025	+0.010	+0.013

Only focal competes, and the pattern is dataset-driven: focal beats TraLO’s macro-F1 on **Derm** across all three backbones (MNV3 −0.012, ViT −0.020, RegNet a tie +0.006), while on Oct/Tissue it ties-to-loses (Oct/MNV3 +0.025; ViT ≈ 0). cc-F1 tied (ceiling); native satisfaction TraLO 1.0 vs every baseline 0.0. **Verdict: honest tie on quality + native satisfaction** – TraLO’s durable, backbone-independent edge is cap satisfaction, not a macro-F1 win over focal.

B2 Does the tight-cap advantage reproduce at native (224px) resolution?

192 runs, 5 native datasets × warm-up{1,5} × L30(+L20) × 4 seeds. Compared to the best *trained* baseline; cc-F1, mean over seeds.

dataset	TraLO	best baseline [†]	Δ = TraLO – best
Derm	0.331 / 0.412	0.325 / 0.412	+0.006 / −0.000
Retina	0.181 / 0.208	0.181 / 0.200	+0.000 / +0.008
TissueNat	0.212 / 0.265	0.211 / 0.254	+0.001 / +0.011
OrganA	0.395 / 0.463	0.392 / 0.463	+0.003 / +0.000
Blood	0.460 / 0.457	0.460 / 0.457	+0.000 / +0.000

[†] best *trained* baseline (Fioretto-LDF); each cell is warm-up 1 / 5. Post-hoc clip excluded here – at warm-up 1/5 it is undertrained, so a gap over it is not meaningful; Hounie was not run natively. *At native resolution TraLO ties the best trained baseline on all 5 datasets and both warm-ups* ($|\Delta| \leq 0.011$). *An honest negative replication: the tight-cap cc-F1 advantage does not appear at native res – consistent with the mechanism (both trained methods re-rank equally when CE has headroom). TraLO still satisfies the cap natively* (1.0 vs clip 0.0).

B3 Does ALM (the standard windup fix) close the gap? — headline result

24 runs: Fioretto-LDF+ALM update, OctMNIST L30/L40 × MNV3/RegNet/ViT × 4 seeds; cc-F1 gap, 6 cells each. ALM is the *best trained baseline* (it beats plain Fioretto).

comparison	mean	W/T/L	MNV3 L30/L40	RegNet L30/L40	ViT L30/L40
TraLO – ALM (best base.)	+0.028	6/0/0	+0.009/+0.013	+0.048/+0.008	+0.046/+0.046
TraLO – Fioretto	+0.039	6/0/0	+0.016/+0.022	+0.035/+0.021	+0.086/+0.051
ALM – Fioretto	+0.010	5/0/1	+0.007/+0.009	−0.013/+0.013	+0.041/+0.005

Raw: **TraLO 0.455 > ALM 0.427 > Fioretto 0.417**. ALM is the strongest dual (beats Fioretto +0.010) yet **TraLO tops it in all 6 cells** (+0.028). ALM does not close the gap ⇒ the A12 justification stands. Strongest new result.

B4 Are the components (reset/hinge/ρ/freeze) load-bearing in a tie region?

20 runs: DermMNIST L50 (a tie cell), MNV3, leave-one-out × 4 seeds; deltas full – variant.

removed	macro-F1 Δ	cc-F1 Δ	removed	macro-F1 Δ	cc-F1 Δ
–reset	+0.002	+0.006	– ρ -sched	+0.002	+0.003
–hinge	–0.003	0.000	–freeze	–0.002	+0.003

No component load-bearing in a tie region (all ± 0.006) $\Rightarrow$ the portable components carry the **tight-cap** advantage specifically – consistent with the mechanism. Answers reviewer R1-M4.

B5 Is the OctMNIST regime classification stable across win thresholds τ ?

No new runs; W/T/L across backbone cells by $\text{cap} \times \tau \in \{.003, .005, .010\}$. Left = vs the best trained baselines (duals); right = vs post-hoc clip (separate category).

vs. trained base.	cap	.003	.005	.010	vs. clip (sep.)	cap	.003	.005	.010
TraLO – Fioretto	L30	4/0/0	4/0/0	4/0/0	TraLO – clip	L40	3/0/1	2/1/1	1/2/1
	L40	4/0/0	4/0/0	4/0/0		L30	2/0/2	2/0/2	0/2/2
TraLO – ALM	L30	3/0/0	3/0/0	2/1/0		L10–20	tie	tie	tie

vs the trained baselines: the win at L30/L40 is **stable at every τ** (4/0/0 vs Fioretto even at $\tau=.010$). vs post-hoc clip: **never** a stable win – the honest cc-F1 tie. Regime classification is robust.

B6 Bootstrap 95% CIs on the OctMNIST headline (does the win survive $n=4$ seeds?)

No new runs; grand-mean cc-F1 gap, 20 000-resample CI over backbone cells per cap. Left = vs trained baselines; right = vs post-hoc clip (separate category).

vs. trained	cap	mean	95% CI		vs. clip (sep.)	cap	mean	95% CI	
TraLO – Fio	L30	+0.046	[+.024, +.074]	>0	TraLO – clip	L10	+0.001	[–.005, +.007]	0
	L40	+0.033	[+.021, +.044]	>0		L15	+0.004	[.000, +.007]	0
	L20	+0.002	[.000, +.005]	tie		L20	+0.001	[–.008, +.007]	0
TraLO – ALM	L30	+0.035	[+.010, +.048]	>0		L30	–0.003	[–.013, +.008]	0
	L40	+0.022	[+.008, +.046]	>0		L40	+0.002	[–.010, +.012]	0

vs the trained baselines: CIs **exclude 0** at L30/L40 (the flagged ViT L30 cell is +0.086 vs Fioretto – survives). vs post-hoc clip: CI **includes 0 at every cap** (cc-F1 tie). The win is a **mid-cap (L30/L40)** phenomenon – tightest caps ceiling-crush every method. **B7 (BH-FDR, $q=.05$): 3/8 cells survive vs Fioretto, 0/8 vs clip.**

B8 Does the cc-F1 win reproduce on a fourth backbone (MobileNetV2)?

32 runs: MNV2 $\times$ OctMNIST L30/L40 $\times$ {TraLO, Fioretto, Hounie, clip} $\times$ 4 seeds; cc-F1 gap.

comparison	mean	L30	L40	raw cc-F1
TraLO – Fioretto (best trained base.)	+0.042	+0.047	+0.037	TraLO 0.454,
TraLO – Hounie (collapses)	+0.169	+0.189	+0.149	Fioretto 0.412,
TraLO – clip (separate category)	+0.007	+0.009	+0.005	Hounie 0.285, clip 0.447

Reproduces on a 4th backbone: TraLO beats the **best trained baseline** (Fioretto) by +0.042; Hounie collapses. vs post-hoc clip: a cc-F1 tie (+0.007) but clip has 0.0 native satisfaction. **Backbone-general win vs the trained baselines** across MNV3/RegNet/ViT/MNV2.

Gaps vs the literal spec & the mechanism

(1) **B1 ViT-B/16 – done.** The spec wanted 3 backbones; all 60 ViT runs are complete (0 failed) and folded into B1. The three-backbone picture is dataset-driven (focal wins Derm on every backbone, ties/loses on Oct/Tissue), confirming the tie-on-quality verdict. **Gap closed.** (2) **B2 warm-up-50 native replication – declined.** We ran short warm-up to isolate the mechanism, which + B5–7 already predict the warm-up-50 native result is a tie vs the trained baseline. (3) **Full 3-dataset regime table + explicit per-cell bootstrap / q -values** – regenerable from the existing corpus, no new runs.

Mechanism (the spine). The count penalty reaches the weights only through a scalar soft-count, so its gradient is a *scalar* $\times$ *fixed direction* and its scale (λ, ρ) is absorbed by Adam + gradient clipping. With CE **headroom** and a binding tight cap it *re-ranks* the boundary $\rightarrow$ real cc-F1 gain over the trained duals (B3, B8); once CE **saturates** (loose cap / native res / easy data) it only *shifts* logits uniformly $\rightarrow$ same top- $K \rightarrow$ tie (B5–6, B2). Post-hoc clip is a separate axis: it can match cc-F1 but cannot satisfy the cap natively. This single mechanism predicts every Track B win and tie.